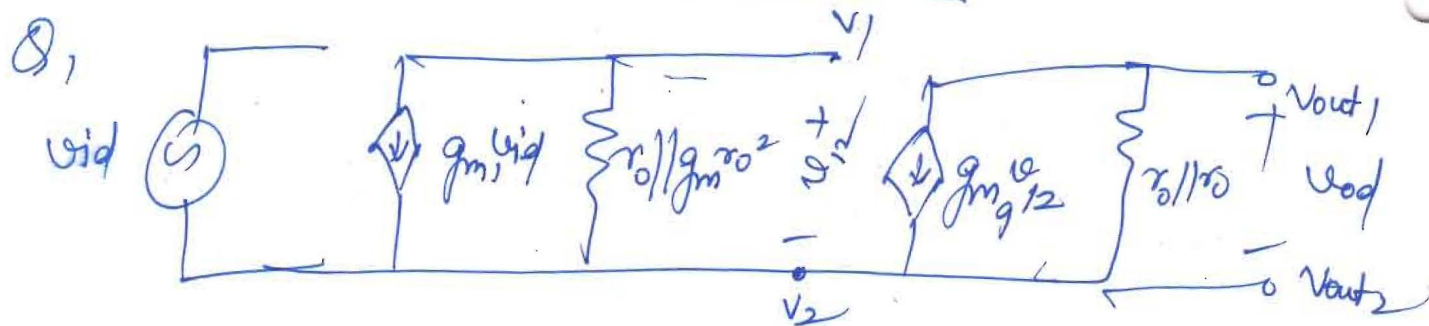


PART B Sol



$$A_v(\text{max.}) = 5000$$

$$\text{Max input (for undistorted o/p)} = \frac{V_{out \text{ swing}}}{A_v} = 4.6 \text{ mV p-p}$$

$$\omega_{out} \Rightarrow 0.2 \text{ M rad/sec} = \omega_{-3dB}$$

$$\omega_{1,2} = 1 \text{ M rad/sec}$$

$$\omega_{GB} = 1 \text{ G rad/sec} = A_v \omega_{-3dB}$$

Shift ω_{GB} to 1 M rad/sec

$$\text{New } \omega_{-3dB} (\text{reqd.}) = 0.2 \times 10^3 \text{ rad/sec}$$

$$\text{So } C_{\text{additional}} = 99.9 \text{ nF}$$

$$C_i = C_L$$

Q.1.

$$Z_{out} = (S_1 \text{ open}) = \frac{1}{sC_L}$$

$$Z_{out} (S_1 \text{ closed}) = \frac{1}{sC_L} \parallel (R_1 + \frac{1}{sC_1})$$

$$= \frac{1}{s(2C_L)} \frac{(1 + sR_1C_L)}{(1 + s\frac{R_1C_1}{2})}$$

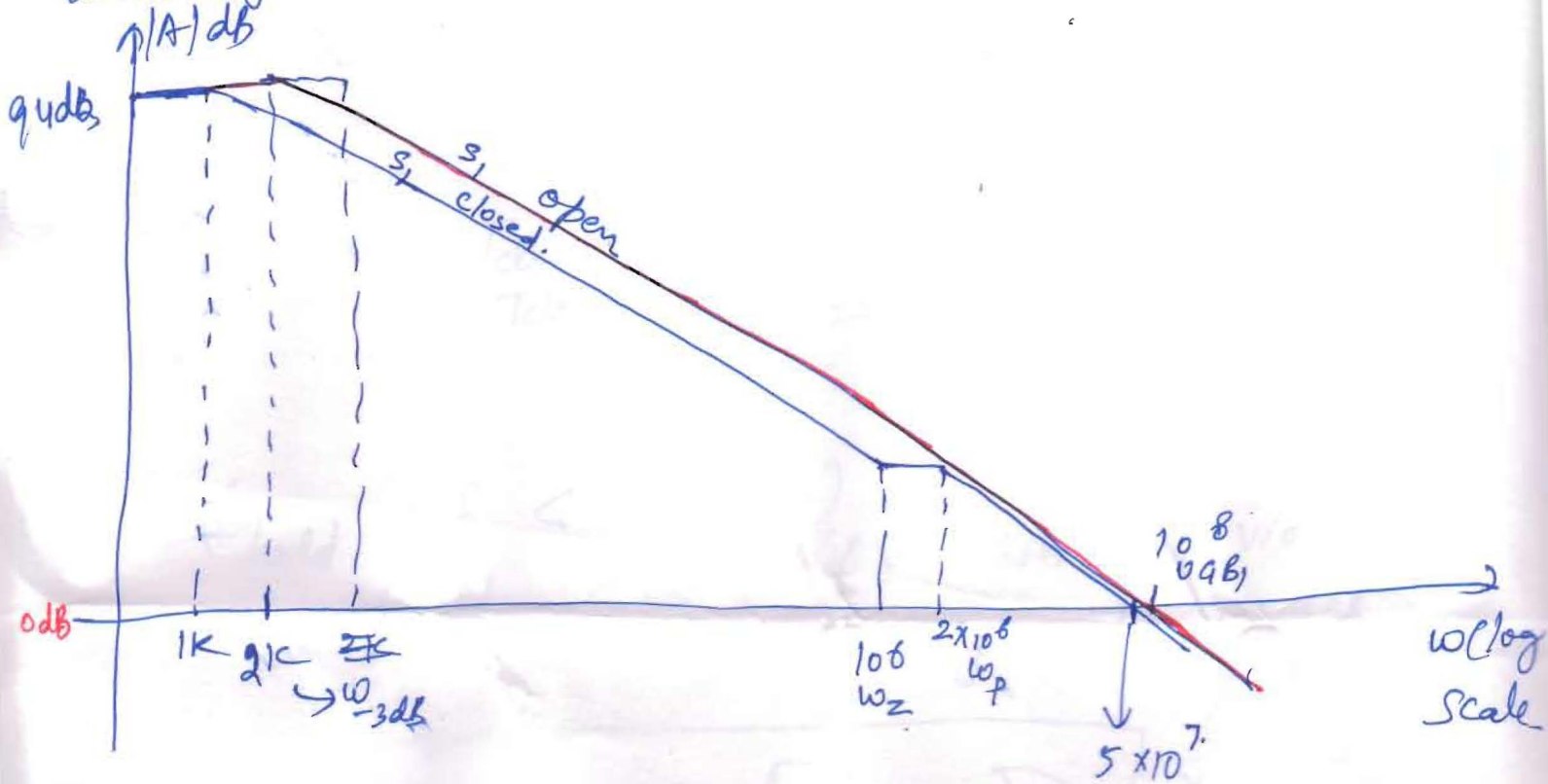
$$A_0 = 50,000 = (94 \text{ dB})$$

$$\omega_{out}(S_1 \text{ open}) = 2 \text{ K}, \quad \omega_{out}(S_1 \text{ closed}) = 1 \text{ K}$$

$$\omega_z = (1 + s R_1 C_1) = 10^6 \text{ rad/sec}$$

$$\omega_p = (1 + s \frac{R_1 C_L}{2}) = 2 \times 10^6 \text{ rad/sec}$$

Bode Mag. plot (corner)



Q.2

Q.
(a)

Path

A → = 44 ns delay max.

P → = 43 ns (max)

(C →)₁ = 45 ns (max)

(C →)₂ = 22 ns (max)

$t_c = 10 \text{ ns}$

(b) T_{clk} required

$$T_{clk} + \delta \geq T_{C-Q} + t_{logic} + t_{su}$$

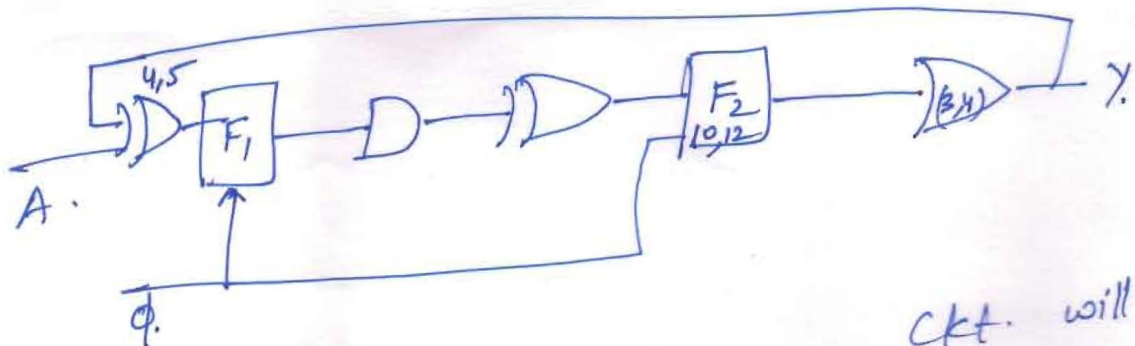
$$T_{clk} + 9 \text{ ns} \geq (9 + 14 + 8) \text{ ns}$$

$$T_{clk} \geq 22 \text{ ns} \Rightarrow f \geq 45 \text{ MHz}$$

(c) $t_{hold} + \delta < 15 \text{ ns}$ (min)

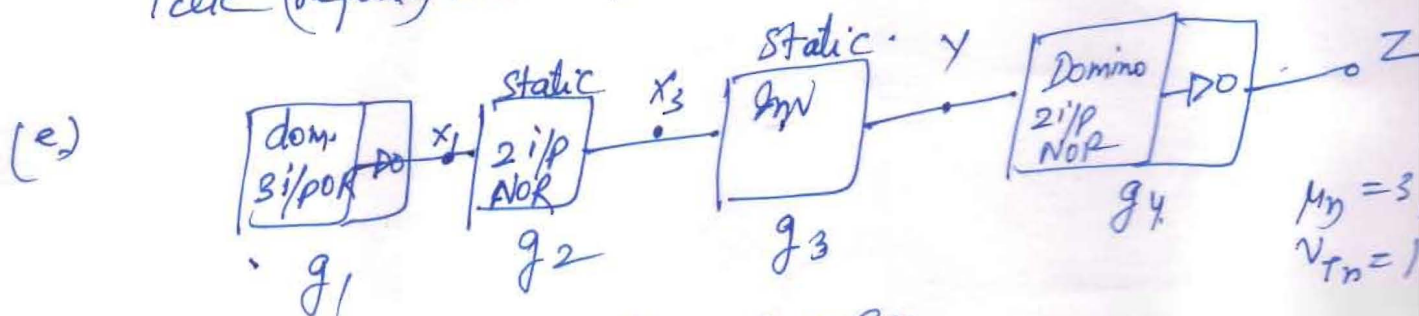
$t_{hold} < 6 \text{ ns}$

(Yes hold time violation)
given $t_{hold} = 8 \text{ ns}$



Ckt. will not work

T_{clk} (required) now $\geq 29 \text{ ns}$



$$\text{Total } G = g_1 \times g_2 \times g_3 \times g_4$$

$$= \frac{2}{4} \times \frac{7}{4} \times 1 \times \frac{2}{4} = \frac{28}{64}$$

$M_n = 3$
 $V_{Tn} = 1$